Design, Construction and Efficiency Evaluation of a 220 V/ 35 V & 4A single Phase Transformer

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***Abstract*—This project presents the design, construction, and performance evaluation of a single-phase step-down transformer rated at 220 V / 35 V, 4 A for low-voltage AC applications. The transformer was fabricated using an EI-laminated core, 24-SWG primary winding, and 17-SWG secondary winding. A complete theoretical design was carried out, including core area calculation, turns-per-volt estimation, and determination of primary and secondary turns. The construction process involved core preparation, winding, insulation, and final assembly, with careful attention to electrical and mechanical reliability. To evaluate performance, standard open-circuit (OC) and short-circuit (SC) tests were performed. The no-load test provided core loss characteristics, yielding a measured secondary no-load voltage of 35.76 V, which closely matches the theoretical design value. The short-circuit test allowed estimation of copper losses and equivalent impedance, with a measured copper loss of 25 W. At full load, the transformer delivered an output power of 140 W with an efficiency of approximately 80%. The results demonstrate acceptable voltage regulation, stable operation, and reasonable efficiency for a small laboratory transformer. The study confirms that properly constructed low-power transformers using appropriate wire gauges, insulation, and laminated EI-cores can achieve performance comparable to typical commercial units.**

Keywords— Transformer design; open-circuit test; short-circuit test; core loss; copper loss; voltage regulation; efficiency; EI-core transformer; low-voltage supply.

I. INTRODUCTION (*HEADING 1*)

Transformers are electromechanical devices that transfer electrical energy between circuits via electromagnetic induction, enabling voltage conversion, electrical isolation, and efficient power distribution. Small single-phase step-down transformers are commonly used in domestic, laboratory, and low-voltage industrial applications.

This project involves designing and constructing a single-phase transformer rated 220 V to 35 V, 4 A, focusing on winding techniques, core selection, insulation, and performance testing. A laminated EI-core with 24-SWG primary and 17-SWG secondary enamel-coated copper windings was used. Standard tests, including open-circuit (OC) and short-circuit (SC), were performed to determine core loss, copper loss, equivalent circuit parameters, and efficiency. The measured no-load voltage of 35.76 V closely matched theoretical values, confirming the design's accuracy.

The results show that a well-designed small transformer achieves good voltage regulation, acceptable losses, and reliable efficiency, while providing practical understanding of magnetic circuits, winding techniques, and transformer characterization.

II. DESIGN METHODOLOGY

A. *Theoretical calculation*

The design of the transformer is based on standard transformer equations and practical engineering considerations. The design requirements are:

- Primary Voltage: 220 V AC
- Desire output Voltage = 35 V AC
- Output Current = 4A
- Frequency = 50 Hz
- Output Power, P = V*I =140W

Input Power ≈ Output Power

So, Primary Current=140/220=0.64A

Step 1: Wire selection

- Primary winding: 24 SWG (rated ≈ 0.8 A)
- Secondary winding: 17 SWG (rated ≈ 4.1 A)

These gauges satisfy safe thermal and conduction limits.

Step 2: Core Area selection

Total power =140 W

Approximate core area:

$$A_c = \sqrt{P} = \sqrt{140} \approx 11.8 \text{ cm}^2$$

From table Formwork = Ac=12.8cm2(3.2cm×4cm)

suitable for a 150 W class transformer.

Step 3: Number of Turns

Turn per volt, T.P.V =42/Area of the core

=42/12.8

=3.28 T/V

For 220V, Number of turns at primary side-N1 = 220*3.28 = 721.6 = 722

For 35V, Number of turns at secondary side-N2= 35×3.28=114.8≈115 turns

B. Major specification of Material and Component

Core Materials

- Core Type: Laminated EI-core
- Core Material: CRGO (Grain-Oriented Silicon Steel)
- Lamination Thickness: 0.35 mm
- Core Cross-sectional Area: 12.8 cm² (3.2 cm × 4 cm)

Winding Materials

- Primary Winding Wire: 24 SWG enamel-coated copper
- Secondary Winding Wire: 17 SWG enamel-coated copper
- Conductor Insulation: Super-enamel, Class B (130°C)

Insulation Materials

- Interlayer Insulation: Polyester or kraft transformer paper
- Primary–Secondary Isolation: Double paper insulation
- Varnish: Heat-resistant insulating varnish (air-drying)

Structural Components

- Bobbin Material: Transformer-grade thermoplastic bobbin
- Clamping Structure: Mild-steel plates and bolts
- Fastening Components: Mild-steel clamping plates and bolts

III. Design Diagram

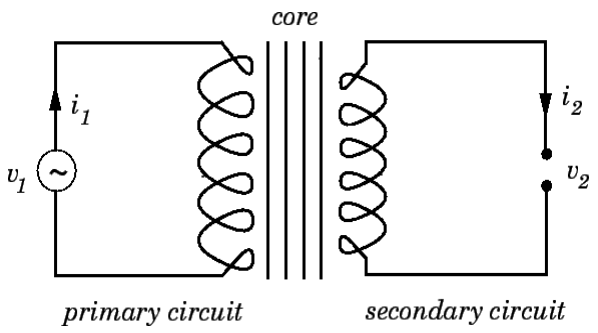


Fig1: Circuit Schematic of transformer

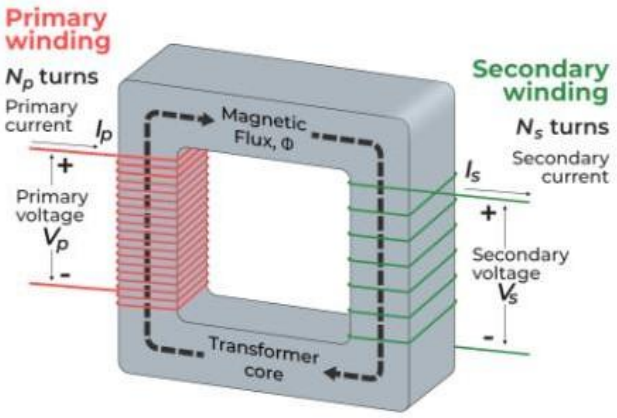


Fig2: Schematic Diagram of Transformer

IV. Physical Construction Process

A. Primary winding construction

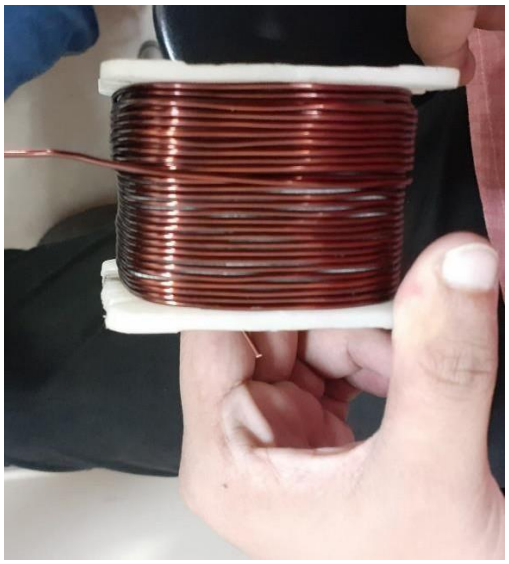
The primary winding was designed for a 220 V AC input and required 722 turns of 24 SWG enamel-coated copper wire. Before beginning the winding process, the bobbin was carefully wrapped with multiple layers of insulating kraft paper to ensure proper electrical isolation from the magnetic core.



The primary coil was wound uniformly, layer by layer, with consistent tension applied throughout the process to maintain compactness and avoid air gaps within the winding. Each completed layer was separated using interlayer insulation to prevent turn-to-turn breakdown and to ensure long-term thermal and electrical reliability.

B. Secondary winding construction

The secondary winding was designed for a 35 V, 4 A output and consisted of 105 turns of 17 SWG enamel-coated copper wire. Before starting the secondary winding, a fresh and complete layer of high-quality insulation paper was applied over the finished primary winding to ensure proper electrical separation and to meet safety requirements.

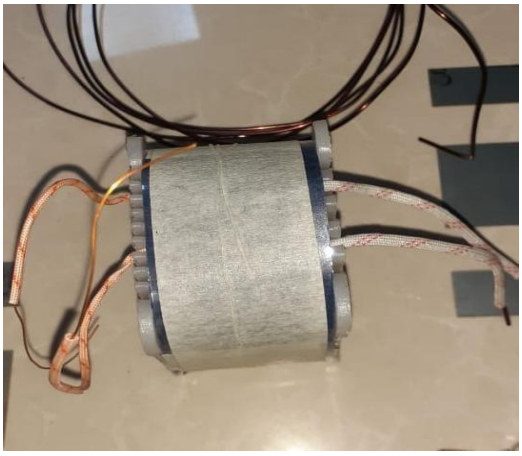


The secondary coil was wound using thicker 17 SWG wire to safely carry the higher output current and minimize copper losses. The winding was completed with uniform tension and proper alignment to maintain structural stability, reduce resistive heating, and ensure efficient power delivery under load.

C. Final Insulation

After completing the secondary winding, the entire coil assembly underwent a **final insulation process** to ensure electrical safety and reliable long-term operation. A layer of high-grade transformer insulation paper was wrapped around the outer surface of the secondary winding, fully covering all exposed copper and evening out the winding profile.

This final insulation layer provides adequate dielectric strength, prevents direct contact between the windings and the EI-core laminations, and significantly reduces the risk of electrical faults such as short circuits, tracking, or insulation breakdown during operation. A coating of insulating varnish was also applied to enhance thermal stability and mechanical rigidity.



D. Assembly of core and coil

Once both the primary and secondary windings were completed and fully insulated, the bobbin assembly was inserted into the EI-core window. The **E and I laminations** were then reassembled carefully, ensuring minimal air gaps to maintain optimal magnetic performance. The laminations were stacked tightly and secured using steel clamps and bolts to reduce vibration, improve magnetic coupling, and enhance the structural stability of the transformer.



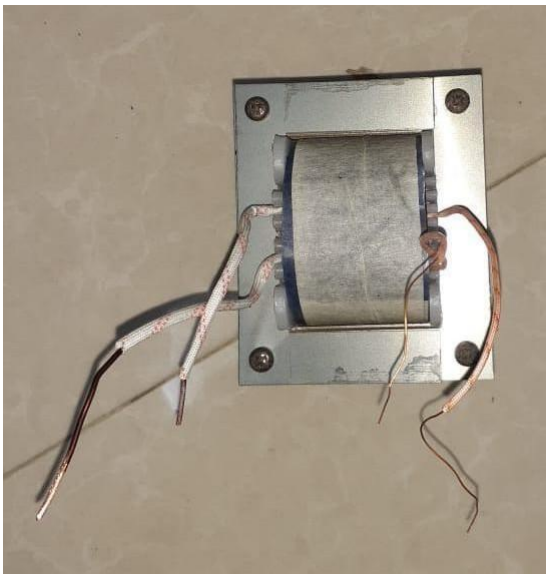


Fig: Final Transformer

V. Performance Testing and Analysis

A. No-load Voltage Test

- Supply Voltage 220 V AC at Primary.
- Measured 29.87 V at Secondary.



Fig: Output Voltage

B. Open Circuit Test

- Applied Voltage 30 V AC
- No Load current 0.2A
- Measured Secondary Voltage: 35.76 V
- Core (Iron) Loss: 3 W (The input power is measured using a wattmeter)

C. Short Circuit Test

- Applied Voltage: 30V
- Rated Current of Primary side: 0.91 A
- Copper loss: 25W

D. Efficient of transformer

At full load,

- Input power 140 W.
- Total loss = 3W + 25W = 28W.
- Output Power = 140 W – 28 W = 112 W.

Efficiency = (112/140)*100%

=80%

CONCLUSION

This project involved designing, building, and testing a 220 V to 35 V, 4 A single-phase transformer. Key design steps included calculating turns, choosing wire sizes, estimating core area, and applying proper insulation. Careful construction ensured good alignment, insulation, and mechanical stability.

Open-circuit and short-circuit tests confirmed the design accuracy. The measured no-load voltage of 35.76 V closely matched the expected output, and the short-circuit test provided reliable copper-loss and resistance data. The transformer achieved an efficiency of about 80% at full load.

Overall, the results validated the design and showed that small EI-core transformers can perform well when properly constructed. The project strengthened understanding of transformer theory, magnetic circuits, and standard testing methods.

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